What to show the operator of a free-floating arm: reach maps against rehearsal under a base tilt budget

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Abstract

An arm on a free-floating spacecraft turns the spacecraft when it moves. If the base must stay within a tilt budget, here 2 degrees, only part of the arm's reach can be used, and an operator needs to know which part. That the reach of such an arm depends on the path is known since the early 1990s, and a planning aid that previews a move and checks it before it is executed was proposed in 1992. Reachable sets under a limit on base attitude have been built for a 3 joint arm. What we did not find measured is how far a map drawn for the operator can be trusted under an attitude budget, and we found no study with people. We measure it in simulation for a 7 joint arm under five controllers. One move away from rest, a map pulled in to 0.8 of its length holds from all of 10 starts under the two controllers that ignore the base, and from none of 10 under the three reactionless ones, which reach further (median 0.45 to 0.46 m against 0.37 to 0.39 m). Over sessions of 40 moves no controller keeps it: the whole map still holds in 0 to 4 of 10 sessions, and three sessions end stuck, all under controllers with no pull back to a rest posture. A map recomputed from the current state is right and 12 to 13 s late. A cautious map looked up from 600 stored states needs no simulation and keeps 0.67 of the reach, against 0.29 for one fixed map made as cautious, but it still promises 0.05 m too much on 1.4% of lines, in 43 of 199 test states. The rehearsal of the command itself is the reference for all of this, and in two fenced runs it disagreed with what the arm then did on 1 of 45 and 2 of 46 goals. We describe a tool that runs four displays (the rest map, the map that held, the map from the current state, no map) and a within-subject study protocol that compares them. [study results: no participant has been run]

Keywords: free-floating manipulator, teleoperation, workspace, base attitude, operator feedback

1 Introduction

A manipulator on a free-floating base has no fixed workspace in the usual sense. Momentum is conserved, so every joint motion moves the base, and where the hand ends up depends on the path the joints took [1, 2]. Papadopoulos and Dubowsky [3] made this exact: some points can be reached by some paths only. Reaction null-space control [6, 7] removes the base motion when the arm has joints to spare, but only for part of the reach, and the rest costs base attitude.

In practice the attitude has a budget: an antenna has to stay pointed, a camera has to keep its target. So the question an operator actually has is not "can the arm reach that" but "can the arm reach that with the base still inside its budget". The answer is a region, and the natural aid is to draw it.

We do not claim that this region moves. That follows from path dependence and is thirty years old. We also do not claim the idea of checking a move before making it: Papadopoulos [12] proposed a planner that previews the motion and asks the operator to approve it. What is missing is the size of the problem and what to do about it for the person at the controls. How wrong is a drawn map, under which controller, after how many moves, and is a map worth showing at all next to a check of the command itself? Our contributions:

1. The reach of a free-floating arm under a budget on base tilt, with joint limits and collision with the bus in it, measured for a 7 joint arm under five controllers. A tilt limit during teleoperation is not new: Sato et al. [13] kept one in 1993 by slowing the arm, on a satellite with attitude control. With nothing holding the base, the limit turns into an edge of the reach, and that edge is what we measure.
2. A measured comparison of what can be shown: the map from rest, the same map pulled in, a map recomputed from the current state, a map looked up from stored states, and the rehearsal alone. One move from rest and over sessions of 12 and 40 moves.
3. A finding we did not expect: the map fails first under the controllers built to keep the base still. They reach further, and their failed moves end with joints on their limits, which we counted and did not explain. Plain controllers reach less and keep a pulled-in map for one move, then lose it over a session.
4. A teleoperation tool and a study protocol, written before any data, that compare the displays with people. [and the result]

2 Related work

Workspaces of free-floating arms. Umetani and Yoshida [1] give the generalized Jacobian and already separate a workspace reachable along a straight path from a given posture from one that is guaranteed. Papadopoulos and Dubowsky [3] show that singularities depend on the mass distribution and on the path taken, and define a path-dependent workspace and a path-independent one inside it, where any path works. Vafa and Dubowsky [4] give the virtual manipulator. In all of these the workspace is bounded by singularity or by link lengths, and the base is free to turn. Three lines of work do limit the base. Umetani and Yoshida [19] define five workspaces for a

space arm, one with the attitude constrained and one guaranteed. Jia et al. [18] build reachable sets for a 3 joint arm, among them one that respects a limit on how far the base attitude may deflect. Cocuzza et al. [20] define the zero reaction workspace, where the arm moves with no reaction on the base at all, and later work takes it to arms with many joints [21]. Ours can be read as that workspace with the zero relaxed to a budget. What we add is not the set. It is what happens to the set when it is drawn for an operator: on a 7 joint arm with joint limits, under five controllers, over a session. We know these three lines from abstracts and summaries only. [read Jia 2019, Umetani and Yoshida 2001 and the zero reaction workspace papers in full] For a planar two-link arm we checked our own code against this theory: the band of radii where dynamic singularities exist comes out at 0.41 to 2.36 m on a grid and at the same two numbers from the masses alone (Section 4.5).

A planner for the operator. The closest work to ours is Papadopoulos [12]. The operator names a target and a path. The planner checks whether the path passes through or near a dynamic singularity, previews the simulated motion, and executes after approval. If the path fails it proposes another, built from a detour through the path-independent workspace and small closed motions of the hand that turn the base on purpose. Our rehearsal is this check with a different test in it. The differences are these. The 1992 planner tests for singularity, ours for a tilt budget, joint limits and contact with the bus (collision is named there and left out). It turns the base to get somewhere, we are not allowed to. The example is a planar arm with two joints, so there is no redundancy and no reactionless control. It draws no map, so it does not ask whether a map could replace the check. No operator was tested. We do not propose alternative paths, which that paper does. We read it except for its example section.

Keeping the base still. Torres and Dubowsky [5] map which joint motions disturb the base least. Reaction null-space control [6] and its flight test on ETS-VII [7] remove the disturbance where redundancy allows. Cocuzza et al. [14] minimise the reaction torque by least squares with joint limits inside the solution. Nenchev and Yoshida [23] compare the pseudoinverse of the generalized Jacobian with a null-space controller that keeps the base still, on a 7 joint arm under velocity teleoperation, and find the pseudoinverse the more reliable (from the abstract). These are controllers. They do not tell an operator where the controller will stop working.

A tilt limit while teleoperating. Sato et al. [13] is the nearest work on the budget itself. A ground operator drives a satellite arm, and the satellite attitude has to stay within set limits so the link to the ground holds. Their controller slows the arm as the predicted disturbance grows and feeds the disturbance forward to the attitude control. The operator works in a graphic simulator and, in their words, does so without considering the disturbance. So the limit is hidden from the operator on purpose, and the satellite has attitude control to recover with, which is why slowing down is enough. Ours has none. Slowing a free-floating arm does not change where its base ends up, so the limit cannot be hidden: some goals are simply out. Tested in a simulator, no operators compared. [read from the abstract only, get the paper]

Drift over repeated moves. A redundant arm that brings its hand back does not bring its joints back, and on a free base the attitude does not come back either. Chen et al. [15] plan motions of a free-floating arm so that hand, joints and base

attitude all reset at the end of a task, and review the earlier work on repeatability of redundant arms. That work removes the drift by planning. We measure what the drift does to a displayed map over a session when the operator picks the goals and nobody plans against it. Our finding that sessions without a pull to rest posture get stuck may be this known drift seen from the operator's seat. We have not shown that it is.

Showing limits to operators. Virtual fixtures [8, 9] put limits in front of users, and have user studies, but the limits are fixed in space. Coelho et al. [16] give the operator of an aerial manipulator a haptic wall at the limits of the null-space motion of the base. That wall moves with the state, the base is actuated, and the test is a numerical simulation with no users. Hauck et al. [24] draw the reachable space for people planning robot motions in virtual reality and test it with 20 of them, for an arm on a fixed base. Nearest to us, Pietras et al. [22] describe ways to support a ground operator against breaking limits on base attitude, such as antenna pointing, on a real-time simulator meant for tests with an operator in the loop. We have the abstract only, and it reports no test with people. [get the Pietras paper: if it tested operators, the last claim of the gap changes]

The gap. Path dependence is known, a check before execution is known, a tilt limit in teleoperation is known, joint drift is known, and reachable sets under a limit on base attitude exist for a 3 joint arm [18]. We found no work that draws such a set for the operator of a base that nothing holds and measures how far it can be trusted across controllers and over a session, and we found no study with people. This is what about 80 searches turned up, with one paper read almost in full and the rest from abstracts and excerpts, not a proof that nobody did it. Japanese and Chinese work and the conference papers of the 1990s are thinly covered. [one systematic search before submission, with forward citations]

3 System

A free-floating bus carries a 7 joint arm in MuJoCo [10]. Nothing holds the bus: no thrusters, no reaction wheels. The hand is driven by resolved-rate control on the generalized Jacobian, and the spare joint is used to cancel base reaction where it can. Base tilt is counted from the attitude at the rest pose, and the budget is 2 degrees. The bus is a 1 m cube of 300 kg. The arm weighs 12.5 kg in seven links and measures 0.97 m from shoulder to hand at full stretch. Joint rates are capped at 2 rad/s. The stacked hand and base rows are inverted with a damping of 10^{-6}, which was never tuned and decides where some lines end (see the paragraph on joint limits inside the controller). [link lengths and gains: table from the CA3 paper]

The reach is probed along 98 directions from the rest position of the hand. Along each one the hand is driven out until the tilt reaches the budget, the arm would hit the bus, or the hand stalls. The 98 end points form the map the operator sees, which we call the rest map. Distance along a direction is given as a fraction of the rest map, so 1 is its edge.

A command is checked by rehearsal: the move is simulated from the state the arm is really in, and refused if it ends outside the budget. This is the check of the 1992 planner [12] with the budget, the joint limits and the bus as the test. Every result in

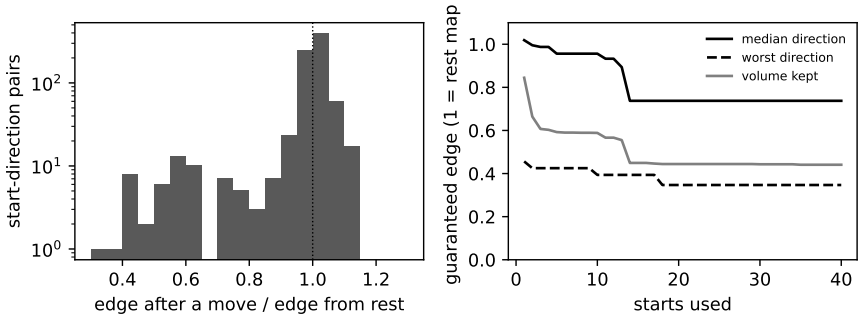


Fig. 1 Left: edge after one move, as a fraction of the edge from rest, 800 start and direction pairs. Most are at 1, a tail collapses. Right: the edge that holds from every start used so far, against the number of starts. Scripted runs.

this paper is scored against it, so here it is right by construction. Against what the arm then did it was off on 1 of 45 and 2 of 46 goals in two fenced runs, and in the stored sessions 4 of 826 accepted moves ended over the budget, by 0.013 degrees at most. It takes about 4 s for a far goal.

4 The edge moves

4.1 Measurement

We moved the hand to a random start one move away from rest, then found by bisection how far it could go along each of 20 random directions before the budget stopped it, and compared with the same search from rest. 40 starts, 800 start and direction pairs.

In 80% of pairs the edge is within 0.05 of the rest map. In 7% it is below 0.8, and the worst is 0.35 (Fig. 1, left). The bad pairs are not a corner of the space: 17 of the 40 starts and 13 of the 20 directions have at least one. They are more common on moves across the start direction (8 to 10% against 3 to 4% along it) and when the start left more tilt behind (15% above 0.8 degrees against 5% below 0.4), but neither tells you which pair will fail.

So for random goals near its edge a map drawn from rest is wrong about one time in fourteen, and neither depth nor tilt left says which one.

4.2 Two explanations that failed

Leftover tilt. The tilt is counted from rest, so a move that leaves tilt behind has less budget for the next one. If that were the cause, the right map would be the rest map with the budget corrected. We repeated the search with the leftover tilt forgiven. The edge did not go back: the median distance to the rest edge was 0.05 with tilt forgiven against 0.03 with it kept. Tilt is a vector and the budget is a ball around the rest attitude, so leftover tilt moves the edge outward in some directions and inward in others.

Enclosed area. For a free-floating system the attitude change over a closed path grows with the area the path encloses. Large shifts do only happen at large area (worst -0.29 m above 0.09 m^2, -0.05 m below 0.03 m^2), but a linear fit on signed area explains 7% of the shift.

We do not have a formula for the edge. What we have is the simulation.

4.3 Whose edge it is

Everything above is under one controller, ours: impedance on the hand with the spare joints used to cancel base reaction. We repeated it under three others: damped least squares and null-space impedance, which ignore the base, and reaction null-space control without impedance. 10 starts one move from rest, each controller against its own rest map on 98 directions.

The two reactionless controllers reach 0.24 to 0.75 m from rest, median 0.45. The two plain ones reach 0.24 to 0.64 m, median 0.37 to 0.39. Pulled in to 0.8, the map of a plain controller holds for every direction from all 10 starts. The map of a reactionless one holds for 90% of start and direction pairs and for every direction from none of the 10 starts. At 0.95 the plain maps hold for 86 to 89% of pairs, so their edge moves too, by less.

A reactionless controller arrives with less tilt spent (median 0.40 degrees against 1.07) and with joints on their limits: its failed moves end with a median of 2 joints at a limit, those of the plain controllers with none. So the far reach is bought with joint range, and what is left of the range depends on the start. This is read off counts, we have not shown the mechanism.

Between 11 and 26 of the 98 lines end in a stall, under every controller. These are not a singularity of the free base. The arm has run out of length. Under our controller all 26 end with the hand 0.96 to 0.97 m from the shoulder, where full stretch is 0.97 m, and with elbow and wrist within 12 degrees of straight. A bus ten or a thousand times heavier leaves the stall where it is, a hand point 0.10 m further out moves it by 0.10 m, and the other controllers stop at the same distance. The reach on these lines is 0.24 to 0.53 m, median 0.29: in those directions the rest pose is close to full stretch. Two things blur this. The last 2 to 14 cm are a crawl the controller makes, because near full stretch it asks for 5 to 15 rad/s and is scaled down to the 2 rad/s cap. And not every stall is this one: on 10 of the 26 lines one joint also sits on its limit, and from two other states 4 of 34 stalls were joint-limit stalls with the arm bent. The map should tell the two apart and does not yet.

Joint limits inside the controller. The reactionless controllers lose the map with joints sitting on their limits, so we tried one that looks at the limits while it solves, after Cocuzza et al. [14]: our controller with a weighted least-norm solve [17], where a joint heading for its limit costs more the closer it gets. It changed almost nothing. Rest reach 0.24 to 0.75 m, median 0.46, and two of the 98 lines move, by 0.08 and 0.09 m. One move out, the 0.8 map holds for 91% of pairs and from 0 of 10 starts, against 90% and 0 of 10 without it. Over 40 moves it ends at 90% of pairs and 0 of 10 sessions, with 23 of 400 moves refused against 15 without it. Our first reason for this was wrong. We wrote that the hand takes three joints, a level base three, and one of seven is left to spare the others with. The count is right and the conclusion is

not. At rest the one free direction is the roll of the tool, which moves neither the hand nor the base and can take no load off another joint. To stop the joint nearest its limit through that direction would need a median of 3400 rad/s against a cap of 2. With the damping in the solve switched off, the weights slow joints 1 to 6 by under 0.1%. What they change at our damping comes through directions the damped inverse has already given up. So the result belongs to that damping: at 10^{-8} the plain controller stalls at 0.40 m on a line where the weighted one reaches 0.73 m, and at our 10^{-6} the two agree to 1 mm. This is one velocity-level version, not their solver, and while the base is held level it has nothing to work with.

We then let the base go a little: the same weights, with the three base rows a cost and not a constraint, and up to 1.5 degrees of the budget to spend on joint room. It is a different trade, not a better one. On the same 392 goals one move out, from 4 starts, it reached 372 and our controller 373. On 89 of the lines it gained 9 goals. On the other 9 its own reach from rest is 10 to 41 cm shorter and it lost 10. Its rest map is smaller, mean reach 0.456 against 0.474 m. Four starts and first values, nothing tuned, and one start carries 6 of the 9.

4.4 Long sessions

An operator does not make one move. We ran sessions of 40 moves, 10 per controller. Each goal is 0.2 to 0.8 of the way to the rest edge along a random direction, rehearsed, and made only if the rehearsal says yes.

In a first run of 12 moves, 10 sessions per controller, the map pulled in to 0.8 holds after move 12 for 91% of directions under our controller, 80% under reaction null space, 93% under damped least squares and, rounded, 100% under null-space impedance. Every direction holds in 0, 0, 4 and 9 of the 10 sessions.

After move 40 the map pulled in to 0.8 holds for 88% of directions under our controller, 83% under reaction null space, 65% under damped least squares and 85% under null-space impedance. Every direction holds in 1, 1, 3 and 4 of the 10 sessions. Null-space impedance held everything in 8 of 10 at move 20, so the loss comes late. Of the 400 goals, all well inside the map, the rehearsal refused 15, 46, 40 and 3. Two sessions under damped least squares and one under reaction null space ended with the last 10 goals all refused. Neither has a pull back to a rest posture. The controllers that have one did not get stuck. Tilt left behind rises over the first moves and levels off (median over the first 10 moves against the last 10: 0.31 and 0.45 degrees for ours, 0.95 and 1.22 for damped least squares). The stuck sessions do not fit that. They sit at a median of 2.01 degrees, on the budget, and all 30 of their refusals are for tilt. A refused move is not made, so nothing brings the base back and the next goal is refused too. Whether drift in posture is what took them there we have not shown.

That a redundant arm drifts in posture over repeated moves is known [15]. What is new here is what it costs the map.

So no controller here keeps a fixed map over a session at 0.8 of its length. The rehearsal is what all of this is scored against, and we expect it to be the thing an operator can lean on. That is a prediction. No operator was tested.

4.5 A planar check

To tie the code to the known theory we built the textbook system separately: a planar arm with two links on a free base, exact momentum, no physics engine, on made-up numbers (base 200 kg, two links of 10 kg and 1 m). With no tilt limit, the radii at which a dynamic singularity exists run from 0.41 to 2.36 m on a grid of joint angles. The same band from the masses and lengths alone, as the sum and difference of the three barycentric lengths, is 0.41 to 2.36 m.

5 What to show instead

5.1 A map that held from every start

Take, for each direction, the worst edge over all starts. After 40 starts the median direction keeps 0.74 of the rest edge and the worst keeps 0.35, which is 44% of the rest reach by volume. The figure was the same at 20 starts as at 40 (Fig. 1, right), so for starts one move from rest it has probably stopped shrinking. It is not a guarantee, and a second move shows it. From 40 starts two moves away from rest, one had already spent the budget on the way. From 24 of the other 39 the rehearsal refused at least one tip of this map, 4.3% of all pairs, on 70 of the 98 directions. Pulling every direction in to 0.8 of its length leaves 11 starts and 0.9% of pairs. A fixed map cannot be made safe by shrinking it against sampled starts. Measured again on the 98 directions of the map itself, same 40 starts, the map that held keeps 0.67 of the rest edge on the median direction, 0.24 on the worst, and 40% of the volume. 28 of the 98 directions keep less than half and 5 keep everything.

5.2 A map from where the arm is

Run the 98 probes again from the current state. On one thread this takes 85 s. Shared over 10 processes it takes 13.0 s from rest and 12.2 s after a detour, on one laptop. So this map is right, for a hand that has stopped, and 12 to 13 s old. The tool shows a yellow marker on the hand while the map is old and a green one when it is current.

5.3 A map looked up

The map from the current state can be stored instead of computed. We took 200 states, one and two moves from rest, and for each the 98 reaches from a fast model without physics. Against the simulation on 199 states that model is 9 mm off on the median, within 0.056 m on 95% of lines and 0.54 m off at worst. A new state is given the weighted average of its 5 nearest stored states. On states left out, the median error is 0.034 m, against 0.120 m for the average of all maps (an average, not the rest map the tool draws). Against the simulated map our first figure came from 5 states: 0.05 m too much on 4.9% of lines. On 101 to 125 simulated states it is 14 to 16% of lines too far by 0.05 m and about 20% too short, median error 0.033 to 0.035 m. So the plain average is too far on about one line in seven, the same failure as the fixed map. It needs no simulation when it answers. We have not timed it.

Made careful. A map that promises too much is the costly kind of wrong, so we made the lookup cautious: per direction the shortest reach among the k nearest stored states, pulled in by a factor, and no reach at all when the base is already over the budget. We scored it against two fixed maps given the same allowance: per direction a low percentile of the stored reaches, and the same shrunk by the share of the tilt budget already spent. The second is the fair one, because the lookup's neighbours carry the tilt for free. 1000 states, one or two moves from rest, 600 stored, 200 to pick the settings on (the furthest-reaching setting that promises 0.05 m too much on under 1% of state and direction pairs), 200 to score, and the scoring states simulated in full. Two of them start with the arm inside the clearance around the bus, where the tool would refuse the move, and are left out.

Against the simulation on 199 states the lookup (10 nearest, pulled in to 0.9) keeps 0.67 of the total reach and promises 0.05 m too much on 1.4% of pairs (0.8 to 2.1 over states). The fixed map (2nd percentile) keeps 0.29 and is over on 0.7% (0.1 to 1.3). The fixed map shrunk with tilt keeps 0.39 and is over on 0.6% (0.3 to 0.9). So the lookup shows 2.3 times the reach of a fixed map and 1.7 times that of the fair one. It does not do so at the same risk. Its misses are spread over 43 of the 199 states, against 9 and 25, and its worst miss is 0.44 m against 0.16 and 0.22 m.

The lookup is only as good as its nearest stored state. The figures in this paragraph are on the fast estimate, not the simulation. Split by distance to the nearest stored state, the furthest quarter of the scoring states is over on 5.0% of pairs, about the same as the fixed map (4.9%), and the nearest quarter on 0.1%. Held out by region, 20 times, the lookup is over on a median of 6.3% against 1.6% for the fixed map, and worse in 19 of the 20. Falling back to the fixed map beyond a set distance halves that and does not close it. So it holds for states drawn like the stored ones and not outside them. Whether a setting keeps the 1% it was picked for is the luck of the split: over 100 random splits the lookup averaged 0.85% and the fixed map 0.75%, each above 1% in about three of ten. With 200 stored states in place of 600 the lookup kept 0.56. A fixed map that holds from every stored state keeps 0.09 of the reach, once the states with no reach at all are set aside.

5.4 No map

Draw nothing. Rehearse every command and refuse the ones that would break the budget. This is as right as the rehearsal and gives the operator nothing to plan with: we expect them to find the edge by being refused.

6 Study

6.1 Design

Within subject, four conditions [decide whether the looked-up map is a fifth]: the rest map (the control), the map that held, the map from the current state, and no map. 20 participants, balanced Latin square over four orders, five per order. Goals come in pairs. The first is easy and moves the arm off rest, and is not scored. The second is the test and starts from where the first left the arm. Whether a goal can be reached

is rehearsed from the arm's real state when it appears. A participant either puts the hand on the goal or gives it up as out of reach, at a cost of 30 s.

Measures, on the second goal of each pair: seconds per goal with the penalty and the rehearsal time in it (primary), commands the fence had to bring back, reachable goals given up, unreachable goals given up, worst tilt, and after each block raw NASA-TLX [11] and one trust rating from 1 to 7.

Hypotheses, written before data: the rest map has the most commands brought back (H1); the map that held has none and the most reachable goals given up (H2); the current-state map is the fastest per goal (H3); no map has no commands brought back but is slower and rated as more work (H4). Friedman test over conditions on the primary measure, Wilcoxon pairs with Holm correction.

6.2 Results

[No participant has been run. Ethics approval and consent to settle first.] A scripted operator has been run through all four blocks, 28 scored goals per display. It is a rule and not a person, and nothing here is participant data. It took 32.6 s per goal with the rest map, 34.7 with the map that held, 34.9 with the map from the current state and 22.1 with no map, and 15 of its sends were refused.

7 Limits

Simulation only. One arm, one bus mass, one budget. The controller comparison and the sessions are 10 starts or 10 sessions per controller. No recovery move was tried on a stuck session. The map that held is tested on one-move starts. The delay of the current-state map is the delay on one laptop. The reach is probed on 98 lines from one point, so the map is a set of lines and not a volume with a proof behind it. The damping in the controller's solve was never tuned and moves where some lines end. The looked-up map holds near its stored states and not outside them. No contact, no flexible parts, no sensor noise.

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